

Classical vs. Modern Methods of System Modelling

Network (Transfer) Function Approach and State-Variable Approach

1. Classical Method: The Network (Transfer) Function

1.1 Definition

The **network function** (transfer function) $N(s)$ of a linear, time-invariant (LTI) system is defined as the ratio of the Laplace transform of the output response $Y(s)$ to the Laplace transform of the input excitation $X(s)$, under the assumption of **zero initial conditions**:

$$N(s) = \left. \frac{Y(s)}{X(s)} \right|_{\text{all initial conditions}=0}$$

It is a purely algebraic (rational) function of the complex frequency variable $s = \sigma + j\omega$, and it completely characterizes the input-output behaviour of an LTI network in the frequency domain.

1.2 Use in Circuit Analysis

- Circuit elements are converted into the **s-domain**: resistor $\rightarrow R$, inductor $\rightarrow sL$ (with initial current as a series source if nonzero), capacitor $\rightarrow 1/sC$ (with initial voltage as a series source if nonzero).
- KVL/KCL equations are written directly in s , converting integro-differential equations into algebraic ones.
- $N(s)$ is obtained by solving these algebraic equations for the desired output over the input.
- Once $N(s)$ is known, the **poles** (roots of the denominator) reveal the natural response (stability, damping, oscillation), and the **zeros** shape the frequency response. Frequency response, step response, and stability can all be read off from the pole-zero pattern without re-solving the differential equation each time.

1.3 Example: Series RLC Circuit

Consider a series RLC circuit driven by input voltage $v_i(t)$, with output taken as the capacitor voltage $v_o(t) = v_C(t)$.

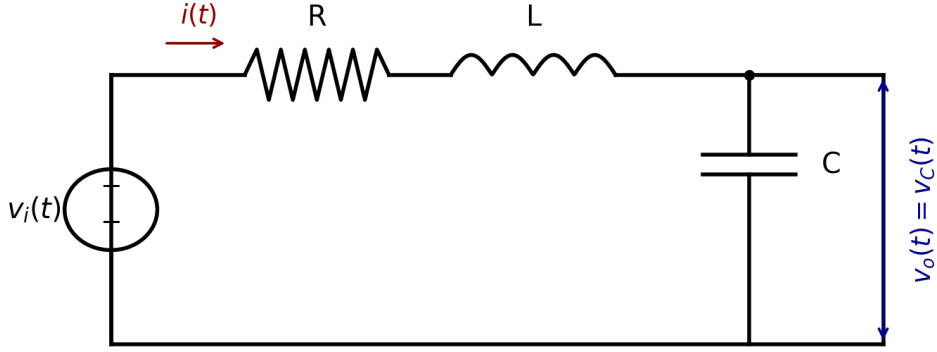


Figure 1: Series RLC circuit with input $v_i(t)$ and output $v_o(t) = v_C(t)$ marked

KVL:

$$v_i(t) = Ri(t) + L \frac{di(t)}{dt} + v_C(t), \quad i(t) = C \frac{dv_C(t)}{dt}$$

Taking Laplace transforms (zero initial conditions) and using $I(s) = sC V_C(s)$:

$$V_i(s) = (R + sL) sC V_C(s) + V_C(s) = V_C(s) [LCs^2 + RCs + 1]$$

$$N(s) = \frac{V_C(s)}{V_i(s)} = \frac{1}{LCs^2 + RCs + 1} = \frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

Comparing with the standard second-order form $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$:

$$\omega_n = \frac{1}{\sqrt{LC}}, \quad \zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$$

Poles of $N(s)$:

$$s_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

1.4 Effect of Varying R: Underdamped vs. Overdamped

- **Underdamped** ($R < 2\sqrt{L/C}$, i.e. $\zeta < 1$): the discriminant is negative, so the poles form a complex-conjugate pair

$$s_{1,2} = -\alpha \pm j\omega_d, \quad \alpha = \frac{R}{2L}, \quad \omega_d = \sqrt{\frac{1}{LC} - \alpha^2}$$

The response is a decaying oscillation (ringing) before settling.

- **Overdamped** ($R > 2\sqrt{L/C}$, i.e. $\zeta > 1$): the discriminant is positive, giving two distinct, real, negative poles s_1, s_2 . The response decays smoothly with no oscillation, more sluggishly than the critical case.

(At $R = 2\sqrt{L/C}$, the poles coincide at $s = -R/2L$ — the critically damped case, the fastest non-oscillatory response.)

1.5 Numerical Problem 1 (Network Function)

Problem: For the series RLC circuit of Fig. 1 with $R = 100 \, \Omega$, $L = 0.1 \, \text{H}$, $C = 100 \, \mu\text{F}$, find $N(s) = V_C(s)/V_i(s)$, its poles, and classify the damping.

Solution:

$$\frac{1}{LC} = \frac{1}{(0.1)(100 \times 10^{-6})} = 1 \times 10^5, \quad \frac{R}{L} = \frac{100}{0.1} = 1000$$

$$N(s) = \frac{1 \times 10^5}{s^2 + 1000s + 1 \times 10^5}$$

Poles:

$$s_{1,2} = \frac{-1000 \pm \sqrt{1000^2 - 4(10^5)}}{2} = \frac{-1000 \pm \sqrt{6 \times 10^5}}{2} = \frac{-1000 \pm 774.6}{2}$$

$$s_1 = -112.7, \quad s_2 = -887.3$$

Check: $2\sqrt{L/C} = 2\sqrt{0.1/100 \times 10^{-6}} = 2\sqrt{1000} \approx 63.25 \, \Omega$. Since $R = 100 \, \Omega > 63.25 \, \Omega$, the two poles are real, distinct, and negative — the circuit is **overdamped**, consistent with the pole values found above.

2. Modern Method: State-Variable Analysis

2.1 Definition of State Variables

The **state variables** of a network are the *smallest* set of variables such that knowledge of their values at $t = t_0$, together with the input for $t \geq t_0$, is sufficient to determine the complete behaviour of the network for all $t \geq t_0$. In electrical networks, the

natural choices are variables associated with energy-storage elements — **inductor currents** and **capacitor voltages** — since these cannot change instantaneously and represent the stored energy of the system.

The general state-variable (state-space) representation of an LTI system is:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad (\text{state equation})$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad (\text{output equation})$$

where $\mathbf{x}$ is the state vector, $\mathbf{u}$ the input vector, $\mathbf{y}$ the output vector, and $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$ are constant matrices for an LTI system.

2.2 Use in Network Analysis

- Write one **first-order** differential equation per energy-storage element using $v = L di/dt$ for inductors and $i = C dv/dt$ for capacitors.
- Express each derivative purely in terms of state variables and inputs — this gives $\mathbf{A}$ and $\mathbf{B}$ directly.
- Any circuit variable of interest is expressed as a linear combination of the states and inputs — this gives $\mathbf{C}$ and $\mathbf{D}$.
- The formulation handles **nonzero initial conditions**, multiple inputs/outputs, and is directly suited to numerical (matrix) computation and simulation.

2.3 Example: State Representation of the Series RLC Circuit

Using the same circuit as Fig. 1, choose state variables $x_1 = i_L(t)$ (inductor current) and $x_2 = v_C(t)$ (capacitor voltage), input $u = v_i(t)$, output $y = v_C(t)$.

From $C dv_C/dt = i_L$:

$$\dot{x}_2 = \frac{1}{C}x_1$$

From $v_i = Ri_L + L di_L/dt + v_C$:

$$\dot{x}_1 = -\frac{R}{L}x_1 - \frac{1}{L}x_2 + \frac{1}{L}u$$

In matrix form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} -R/L & -1/L \\ 1/C & 0 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 1/L \\ 0 \end{bmatrix}}_{\mathbf{B}} u, \quad y = \underbrace{[0 \quad 1]}_{\mathbf{C}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + 0 \cdot u$$

2.4 Eigenvalues of the **A** Matrix

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{bmatrix} -R/L - \lambda & -1/L \\ 1/C & -\lambda \end{bmatrix} = \lambda^2 + \frac{R}{L}\lambda + \frac{1}{LC} = 0$$

$$\lambda_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

This is **identical** to the pole expression obtained from $N(s)$ in Section 1.3. This is not a coincidence: the eigenvalues of **A** always equal the poles of the network function for a state-space realization, since $N(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$ and $\det(s\mathbf{I} - \mathbf{A})$ is the characteristic polynomial in both formulations.

2.5 Numerical Problem 2 (State-Variable Analysis)

Problem: For the same circuit (Fig. 1) with $R = 10 \Omega$, $L = 0.1 \text{ H}$, $C = 100 \mu\text{F}$, write the state-space matrices **A**, **B**, **C**, **D** and find the eigenvalues of **A**.

Solution:

$$\frac{R}{L} = \frac{10}{0.1} = 100, \quad \frac{1}{L} = 10, \quad \frac{1}{C} = 10^4$$

$$\mathbf{A} = \begin{bmatrix} -100 & -10 \\ 10^4 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 10 \\ 0 \end{bmatrix}, \quad \mathbf{C} = [0 \quad 1], \quad \mathbf{D} = 0$$

Characteristic equation:

$$\lambda^2 + 100\lambda + 10^5 = 0$$

$$\lambda_{1,2} = \frac{-100 \pm \sqrt{100^2 - 4 \times 10^5}}{2} = \frac{-100 \pm \sqrt{-3.9 \times 10^5}}{2} = -50 \pm j312.25$$

Since $R = 10 \Omega < 2\sqrt{L/C} \approx 63.25 \Omega$, the eigenvalues are complex conjugates with negative real part — confirming an **underdamped** response, consistent with Section 1.4.

2.6 Numerical Problem 3 (Direct Differential-Equation Model)

Problem: An electrical circuit is represented by the second-order differential equation

$$\ddot{y} + 3\dot{y} + 2y = u$$

where u is the input. Obtain (a) the transfer (network) function and (b) the state-variable representation of the circuit.

Solution (a) — Network function:

Taking the Laplace transform with zero initial conditions ($\ddot{y} \rightarrow s^2 Y(s)$, $\dot{y} \rightarrow sY(s)$):

$$s^2 Y(s) + 3sY(s) + 2Y(s) = U(s) \Rightarrow Y(s) [s^2 + 3s + 2] = U(s)$$

$$N(s) = \frac{Y(s)}{U(s)} = \frac{1}{s^2 + 3s + 2} = \frac{1}{(s+1)(s+2)}$$

Poles: $s_1 = -1$, $s_2 = -2$ — real, distinct, negative $\Rightarrow$ the (unforced) response is a sum of two decaying exponentials e^{-t} and e^{-2t} , i.e. an overdamped-type response with no oscillation.

Solution (b) — State-variable representation:

Since the differential equation is given directly (no explicit circuit elements), the standard choice is the set of **phase variables**:

$$x_1 = y, \quad x_2 = \dot{y}$$

Then:

$$\begin{aligned} \dot{x}_1 &= \dot{y} = x_2 \\ \dot{x}_2 &= \ddot{y} = u - 3\dot{y} - 2y = -2x_1 - 3x_2 + u \end{aligned}$$

In matrix form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\mathbf{B}} u, \quad y = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\mathbf{C}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + 0 \cdot u$$

Check (eigenvalues of $\mathbf{A}$):

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{bmatrix} -\lambda & 1 \\ -2 & -3 - \lambda \end{bmatrix} = \lambda(\lambda+3)+2 = \lambda^2+3\lambda+2 = 0 \Rightarrow \lambda_1 = -1, \lambda_2 = -2$$

As expected, the eigenvalues of $\mathbf{A}$ exactly match the poles of $N(s)$ found in part (a). This example shows that the same $N(s) \leftrightarrow (\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$ correspondence used for the RLC circuit holds for *any* LTI system specified directly by its governing differential equation — the “phase-variable” (or controllable canonical) form used here is a standard, systematic way to go from an n th-order differential equation to a state-space model.

3. Comparison: State-Variable Approach vs. Network Function Approach

Aspect	Network Function (Classical)	State-Variable (Modern)
Initial conditions	Assumes zero initial conditions	Handles arbitrary nonzero initial conditions naturally
System type	Restricted to LTI, SISO (or requires a matrix of transfer functions for MIMO)	Naturally extends to MIMO, and with $\mathbf{A}(t)$, $\mathbf{B}(t)$ to time-varying/nonlinear systems
Internal behaviour	Only input-output relation is captured; internal variables are hidden	Full internal (state) behaviour is available — useful for understanding energy, currents, voltages at internal nodes
Computation	Frequency-domain, algebraic; suited to hand analysis, Bode/Nyquist design	Time-domain, matrix-based; suited to digital/numerical computation and simulation
System properties	Stability inferred from pole locations only	Additionally permits analysis of controllability and observability
Design tools	Classical control design (root locus, frequency response)	Modern control design (pole placement, optimal control, observers)

Summary: The network function method is compact and intuitive for single-input single-output LTI analysis and classical frequency-domain design, but it discards internal structure and cannot handle nonzero initial conditions or multiple inputs/outputs gracefully. The state-variable method retains complete internal information, handles initial conditions and multivariable/nonlinear/time-varying systems naturally, and is the foundation of modern control theory — while still being fully consistent with the network function approach, since the eigenvalues of $\mathbf{A}$ always coincide with the poles of $N(s)$.